

Problem A. Minimum Spanning Tree

Input file: **standard input**
Output file: **standard output**
Time limit: 4 seconds
Memory limit: 256 megabytes

You are given a connected weighted undirected graph G with n vertices and m edges. All vertices are numbered $1, \dots, n$. Your task is to compute the sum of the edge weights in the minimum spanning tree (MST) of G .

Input

The first line contains two positive integers n and m .

Each of the next m lines contains three positive integers u_i, v_i, w_i ($1 \leq v_i, u_i \leq n, 0 \leq w_i \leq 10^9$), describing an edge between vertices u_i and v_i with weight w_i .

Output

Output a single integer: the total weight of the minimum spanning tree of G .

Example

standard input	standard output
7 12 1 2 9 1 5 2 1 6 3 2 3 5 2 6 7 3 4 6 3 7 3 4 5 6 4 7 2 5 6 3 5 7 6 6 7 1	16

Note

$1 \leq n \leq 2 \times 10^5$, $0 \leq m \leq 5 \times 10^5$.

Problem B. Shortest Path

Input file: **standard input**
Output file: **standard output**
Time limit: 2 seconds
Memory limit: 256 megabytes

You are given an undirected graph with n vertices ($1 \leq n \leq 2500$) and m edges ($1 \leq m \leq 6200$). Your task is to find the shortest path from vertex s to vertex t .

Input

The first line contains four integers separated by spaces: n, m, s, t .

Each of the next m lines contains three positive integers s_i, t_i, w_i ($1 \leq w_i \leq 10^9$), describing an edge between vertices s_i and t_i with length w_i .

Output

Output a single integer: the length of the shortest path from s to t . It is guaranteed that at least one path exists.

Example

standard input	standard output
7 11 5 4 2 4 2 1 4 3 7 2 2 3 4 3 5 7 5 7 3 3 6 1 1 6 3 4 2 4 3 5 6 3 7 2 1	7

Problem C. Disjoint Sets

Input file: **standard input**
Output file: **standard output**
Time limit: 4 seconds
Memory limit: 256 megabytes

Given n sets $\{0\}, \{1\}, \dots, \{n-1\}$. At any moment teacher Jingbang gives you two numbers i, j and either asks you to merge sets containing i and j , or asks you whether numbers i and j belong to the same set.

Because teacher Jingbang is demanding, the output uses a special format: for each co-membership question, output 0 or 1 (the answer). Concatenate all co-membership answers from left to right to form a binary string. Interpret this binary string as a binary integer, and output its value mod 998244353.

Input

The first line contains two integers n and m , the initial number of sets and the number of operations. Each of the next m lines contains three integers op, i, j :

- If $op = 0$, merge sets containing i and j
- If $op = 1$, query whether i and j belong to the same set.

Output

Output a single integer: the final answer.

Example

standard input	standard output
3 6 1 1 0 0 0 1 1 0 1 1 1 2 0 2 1 1 2 1	5

Note

$n \leq 4000000$, $m \leq 500000$.

Problem D. Lowest Common Ancestor (LCA)

Input file: `standard input`
Output file: `standard output`
Time limit: 6 seconds
Memory limit: 256 megabytes

Given a rooted tree (a general, not necessarily binary, tree), answer queries asking for the lowest common ancestor (LCA) of two specified nodes.

Input

The first line contains three integers N , M , S — the number of nodes in the tree, the number of queries, and the index of the root node, respectively.

The next $N - 1$ lines each contain two integers x and y , meaning there is an edge directly connecting node x and node y . (The input is guaranteed to form a tree.)

The next M lines each contain two integers a and b , representing a query: find the lowest common ancestor of nodes a and b .

Output

Output M lines. The i -th line should contain a single integer — the LCA of the corresponding query pair.

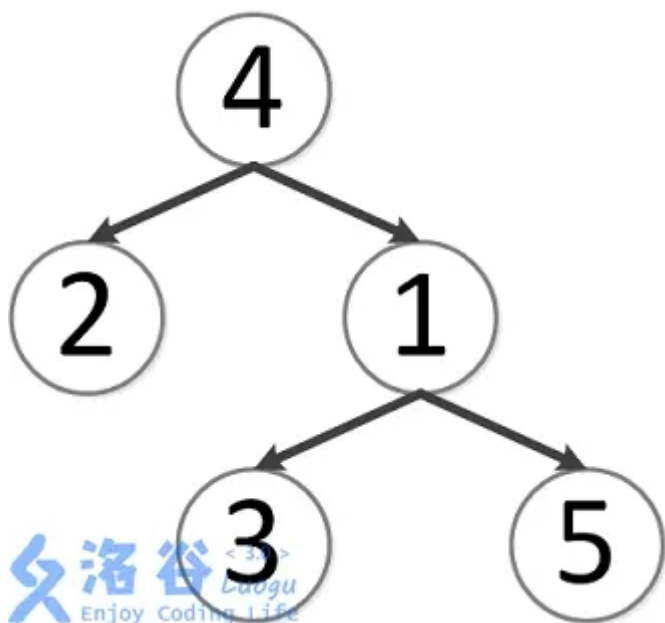
Example

standard input	standard output
5 5 4	4
3 1	4
2 4	1
5 1	4
1 4	4
2 4	
3 2	
3 5	
1 2	
4 5	

Note

$1 \leq N, M \leq 5 \times 10^5$, $1 \leq x, y, a, b \leq N$, and it is not guaranteed that $a \neq b$.

Sample Explanation



The tree structure is as follows:

- Query 1: (2, 4), the LCA is 4.
- Query 2: (3, 2), the LCA is 4.
- Query 3: (3, 5), the LCA is 1.
- Query 4: (1, 2), the LCA is 4.
- Query 5: (4, 5), the LCA is 4.

Thus, the output is: 4, 4, 1, 4, 4.